

Senanayak Sesh Kumar Karri

Research name: K. S. Sesh Kumar

Research Fellow, Brevan Howard Centre for Financial Analysis, Imperial College Business School, London, UK

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Research Summary

Machine-learning researcher working on structured data, time-series analysis, and graph-structured data: non-stationary Gaussian processes on graphs, geometry-aware neural architectures, optimal transport, and submodular and convex optimisation. PhD from INRIA / École Normale Supérieure (SIERRA team) on the links between probabilistic graphical models and submodular optimisation. At Imperial College Business School, current work examines the limitations of machine-learning models in timing the market with limited information such as prices, high-dimensional covariance estimation that combines sectoral graph structure with option-implied forward information to close the gap between expected and realised portfolio risk, and reinforcement learning for portfolio management. Publications at NeurIPS, ICML, JMLR, TMLR, and *Machine Learning*. Two papers accepted at NeurIPS 2026.

Accepted Papers at NeurIPS 2026

- [1] **K. S. Sesh Kumar** and V. Borovitskiy, **Practical Non-Stationary Graph Gaussian Processes**. *Fortieth Conference on Neural Information Processing Systems (NeurIPS 2026)*, accepted as a poster, submission 25512.
- [2] **K. S. Sesh Kumar**, **Geometry-Constrained Kolmogorov–Arnold Networks: Learning Edge Geometry via Banach Duality**. *Fortieth Conference on Neural Information Processing Systems (NeurIPS 2026)*, accepted as a poster, arXiv:2608.25807.

Academic Positions

Machine Learning Research Fellow <i>Brevan Howard Centre for Financial Analysis, Imperial College Business School</i>	London, UK <i>Sept 2022 – present</i>
Research Fellow, Data Science Institute <i>Imperial College London</i>	London, UK <i>Sept 2019 – Aug 2022</i>
Research Fellow, Statistical Machine Learning Group <i>Imperial College London; Leverhulme Centre for the Future of Intelligence fellowship</i>	London, UK <i>Dec 2017 – Aug 2019</i>
Postdoctoral Researcher, Kolmogorov Group <i>Institute of Science and Technology (IST) Austria</i>	Klosterneuburg, Austria <i>Oct 2016 – Oct 2017</i>

Education

PhD in Machine Learning, SIERRA, INRIA / École Normale Supérieure <i>Advisor: Prof. Francis Bach</i> – Thesis: <i>On the Links between Probabilistic Graphical Models and Submodular Optimisation</i> , École Normale Supérieure, 2016.	Paris, France <i>May 2013 – Sept 2016</i>
MSc, Mathematics, Vision, Learning (MVA), ENS Cachan	Cachan, France <i>Sept 2012 – Apr 2013</i>
B.Tech., Computer Science and Engineering, IIIT Hyderabad	Hyderabad, India <i>Jul 1999 – Aug 2003</i>

Selected Publications

- [1] M. Conserva, M. P. Deisenroth, and **K. S. Sesh Kumar**, **The Graph Cut Kernel for Ranked Data**. *Transactions on Machine Learning Research (TMLR)*, 2022.
- [2] C. Buizza, C. Quilodrán Casas, P. Nadler, J. Mack, S. Marrone, Z. Titus, C. Le Cornec, E. Heylen, T. Dur, L. Baca Ruiz, C. Heaney, J. Amador Díaz López, **K. S. Sesh Kumar**, and R. Arcucci, **Data Learning: Integrating Data Assimilation and Machine Learning**. *Journal of Computational Science*, 2021.
- [3] R. Moriconi, M. P. Deisenroth, and **K. S. Sesh Kumar**, **High-Dimensional Bayesian Optimization with Manifold Gaussian Processes**. *Machine Learning*, 2020.
- [4] **K. S. Sesh Kumar**, F. Bach, and T. Pock, **Fast Decomposable Submodular Function Minimization using Constrained Total Variation**. *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.

- [5] R. Moriconi, **K. S. Sesh Kumar**, and M. P. Deisenroth, **High-Dimensional Bayesian Optimization with Projections using Quantile Gaussian Processes**. *Optimization Letters*, 2019.
- [6] **K. S. Sesh Kumar** and M. P. Deisenroth, **Differentially Private Empirical Risk Minimization with Sparsity-Inducing Norms**. *Privacy Preserving Machine Learning (PPML) Workshop, NeurIPS*, 2018.
- [7] **K. S. Sesh Kumar** and F. Bach, **Active-Set Methods for Submodular Minimization Problems**. *Journal of Machine Learning Research (JMLR)*, 2017.
- [8] **K. S. Sesh Kumar** and F. Bach, **Maximizing Submodular Functions using Probabilistic Graphical Models**. *Discrete Optimization for Machine Learning (DISCML) Workshop, NIPS*, 2013.
- [9] **K. S. Sesh Kumar** and F. Bach, **Convex Relaxations for Learning Bounded Treewidth Decomposable Graphs**. *International Conference on Machine Learning (ICML)*, 2013.

Preprints and Working Papers

- [1] F. Allen, M. T. Kacperczyk, and **K. S. Sesh Kumar**, **Adaptive Market Ecology and Conditional Strategy Performance: Evidence from Machine-Learning Market-Timing Strategies**. *SSRN working paper*, 2026.
- [2] F. Allen, M. T. Kacperczyk, and **K. S. Sesh Kumar**, **Modern Machine Learning Tools in Finance: A Critical Perspective**. *SSRN working paper*, 2025.
- [3] S. Cohen, **K. S. Sesh Kumar**, and M. P. Deisenroth, **Sliced Multi-Marginal Monge Optimal Transport**. *arXiv:2102.07115*, 2021.
- [4] **K. S. Sesh Kumar**, A. Barbero, S. Jegelka, S. Sra, and F. Bach, **Convex Optimization for Parallel Energy Minimization**. *arXiv:1503.01563 / HAL 01123492*, 2015.

Awards and Fellowships

- 2019 — Research Fellowship, Data Science Institute, Imperial College London.
- 2018 — Research Fellowship, Leverhulme Centre for the Future of Intelligence.
- 2017 — Data Science Institute research funding, Imperial College London (GBP 20,000).

Earlier Research and Industry Experience

Research Engineer, WILLOW team, INRIA, Paris (2012–2013); Research Intern, SIERRA team, INRIA, Paris (2012); Research Intern, Computer Vision and Geometry Group, ETH Zurich (2011); Research Engineer, Télécom SudParis, Evry (2010–2011); GPGPU Engineer, AMD Research and Development, Hyderabad (2006–2010); Research Assistant, CVIT, IIT Hyderabad (2003–2006), with publications at ICDAR, ICVGIP, DAS, and PReMI.